RACER-GT: A Design-Based Statistical Framework for the Reliable Construction of Long-Horizon Google Trends Time Series

Randomized Acquisition, Global Overlap Calibration, Generalizability Decomposition, and Covariance-Adjusted Consensus

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Abstract

Google Trends (GT) is widely used in finance and economics as a high-frequency proxy for investor attention, information demand, and sentiment. Yet the web interface rescales every request to a 0–100 index, a long daily history must be assembled from several short query windows, and repeating an identical query can return different numbers. A researcher who downloads a series once, stitches the windows sequentially, and feeds the result into a financial model therefore inherits four distinct problems at once: segment-scale error, dependence across repeated retrievals, zero inflation and rounding at low search volume, and attenuation bias in any downstream regression.

This paper develops RACER-GT (Randomized Acquisition, Calibration, Error Decomposition, and Reliability for Google Trends). The framework does not claim to recover Google's absolute search counts. It instead defines two estimands that *are* identified—the expected GT return under a fixed

query and a fixed retrieval design, and the latent relative search signal on a common scale—and then estimates them through eight preregistered stages: (i) a balanced, protocol-locked repeated-retrieval experiment; (ii) global weighted least squares on the overlap graph, which uses every usable overlap simultaneously instead of accumulating error through sequential stitching; (iii) exhaustive retention of exact duplicates together with a residual-based near-duplicate dependence graph; (iv) a three-facet Generalizability Theory decomposition over historical date, collection day, and collection stream; (v) a design-based stratified reference estimator and a covariance-adjusted minimum-variance consensus; (vi) quadratic-programming temporal benchmarking against weekly or monthly aggregates; (vii) separate detection, level, and daily-innovation reliability with a locked acceptance decision tree; and (viii) propagation of the resulting measurement uncertainty into reliability-corrected OLS and SIMEX.

We state the identification conditions explicitly and prove conditional unbiasedness and minimum-variance results under them. Twenty controlled Monte Carlo replications, evaluated at matched information sets, give a mixed result. Cross-pull aggregation cuts mean RMSE against a known latent truth from 7.3519 for a single pull to 3.7421, a 49.1% reduction that improves in all twenty replications, and a further 10.3% below the cross-pull median's 4.1710. Temporal benchmarking contributes about 0.22 more, holding the estimator fixed ($p \approx 3 \times 10^{-12}$). Covariance-adjusted weighting, however, shows no detectable advantage over the simple cross-pull mean of 3.7253 (-0.0167, $p = 0.152$). The natural explanation is that dependence is too homogeneous for GLS to exploit, and we test it directly in a four-scenario experiment that puts a subset of pulls behind a shared cache disturbance cutting across the design facets. The explanation fails: GLS wins no heterogeneous scenario significantly and is significantly worse in two. Diagnostics show the constraints are not responsible (the unrestricted solution is identical to the digit) and that the mechanism works as intended (weights correlate -0.562 with shared residual dependence), leaving the error in estimating $\boldsymbol{\Sigma}$ from nine pulls as what exceeds the efficiency it buys. The theory is untouched — GLS is still BLUE with $\boldsymbol{\Sigma}$ known — but the practical recommendation is not. The calibration stage yields the same shape of result: global overlap-graph calibration does stop error accumulating along the chain of joins relative to sequential stitching (growth ratio 1.12 against 4.75), yet the difference in reconstruction error reaches only $p = 0.098$ under the setting most favourable to it. Taken together these give one cross-cutting conclusion: each mechanism works as designed and can be shown to work, while its effect on point accuracy is second order. The value of this framework is that it

produces a series whose quality can be judged, not a more accurate one. None of these figures is a universal optimality guarantee across keywords, regions, or GT versions.

Keywords: Google Trends; investor attention; measurement error; experimental design; generalizability theory; graph calibration; generalized least squares; temporal benchmarking; SIMEX.

JEL classification: C18, C43, C81, G14.

Contents

1 Introduction

Search data have become a standard empirical input in finance. Da et al. (2011, 2015) showed that Google search frequency for ticker symbols captures retail investor attention in a way that pre-existing proxies do not; Preis et al. (2013) documented associations between search volume for finance-related terms and subsequent market movements; Choi and Varian (2012) established the broader "predicting the present" agenda. A large applied literature now treats a downloaded GT series as if it were an observed variable measured without error.

That treatment is difficult to defend once the retrieval mechanism is examined. Three features of the GT interface interact badly with long-horizon daily research designs.

Per-request renormalization. Every GT response is rescaled so that the maximum value within the requested window equals 100. The returned numbers are therefore not comparable across requests with different windows. Two requests that differ only in their end date produce two different scales for the same calendar day.

Window-length quantization. Daily granularity is returned only for sufficiently short windows. A multi-year daily history must be requested as a sequence of overlapping chunks and reassembled. The conventional reassembly is sequential: chunk 2 is rescaled onto chunk 1, chunk 3 onto the rescaled chunk 2, and so on. Scale errors then compound multiplicatively along the chain, and the final segment inherits the accumulated error of every preceding join.

Non-determinism. Repeating an identical query on a different day, or from a different browsing environment, can return different values. Eichenauer et al. (2022) document this instability directly and propose averaging across repeated samples. Averaging is a substantial improvement over a single download, but it raises a further question that the applied literature has largely left open: repeated retrievals are *not* independent samples, so what exactly does their average estimate, and how much information do N retrievals actually contain?

1.1 What this paper does

We treat data acquisition itself as a designed, auditable measurement experiment rather than as a preliminary inconvenience. The central move is to refuse the question “what is the true search volume?”, which the GT system does not permit us to answer, and to replace it with two questions that are answerable under stated assumptions:

1. Under a fixed query specification and a fixed, preregistered retrieval design, what is the expected value that GT returns for each historical date?
2. On a common scale, what is the latent relative search signal that the repeated retrievals have in common?

The first estimand is design-based and requires no model of Google’s internal sampling; it is unbiased by construction provided the inclusion rule does not depend on the realized values. The second is model-based and requires the assumptions we make explicit in section 3. Reporting both, and keeping them distinct, is a deliberate methodological choice: the design-based quantity is a transparent benchmark against which the model-based reconstruction can be judged.

1.2 Contributions

1. A preregistered acquisition design. Collection day, collection stream, technical replicate, chunk order, and time slot are balanced and locked before any data are seen, and the entire configuration is committed to a SHA-256 protocol hash (section 4). This makes retrieval-side researcher degrees of freedom auditable rather than invisible.
2. Global overlap-graph calibration. We replace sequential stitching with a single weighted least-squares problem on the graph whose nodes are chunks and whose edges are robustly estimated log-scale differences (section 5). This uses all overlap information at once, removes error accumulation, and yields identification, connectivity, conditioning, and cycle-consistency diagnostics that sequential stitching cannot provide.
3. Dependence accounting rather than deduplication. Exact duplicates are retained in the audit archive and collapsed analytically; near-duplicates are identified from residuals after the common signal is removed, so that highly dependent retrievals are not counted as independent evidence (section 6).

4. A three-facet generalizability decomposition. We estimate variance components for historical date, collection day, and collection stream and their interactions, separately for the level, the detection indicator, and the daily innovation (section 7). The associated D-study answers a question practitioners actually face: whether to buy reliability with more collection days, more streams, or more technical replicates.

5. A covariance-adjusted consensus with an honest effective sample size. Weights solve a minimum-variance problem under the estimated residual covariance, and we report spectral and Kish effective pull counts so that dependence is visible in the reported precision (section 9).

6. Propagation into downstream inference. The measurement uncertainty is carried into reliability-corrected OLS and SIMEX rather than discarded at the boundary of the measurement exercise (section 12).

1.3 Scope and what is deliberately not claimed

RACER-GT estimates a latent *relative* search signal on a common scale. It does not estimate absolute Google search counts, and no result in this paper should be read as identifying them. All unbiasedness, consistency, and minimum-variance statements are conditional on the assumptions listed in section 3, and we state in section 14 exactly which guarantees survive which assumption failures.

2 The measurement problem

2.1 Notation

Let $t \in \mathcal{T} = \{1, \ldots, T\}$ index historical calendar dates, and let a *complete pull* i denote one full reconstruction of the historical window under the locked query specification. A complete pull consists of J fixed, overlapping chunks; chunk j covers a contiguous window $W_j \subset \mathcal{T}$, and consecutive chunks overlap in at least m_{min} days.

Write Y_{ijt} for the raw GT value returned for date t in chunk j of pull i. Each pull is characterized by a design cell: a collection day $d \in \mathcal{D}$, a collection stream $s \in \mathcal{S}$, and a technical replicate $r \in \mathcal{R}$.

Remark 2.1 (Streams are composite environments). *A stream is a fixed combination of device, browser profile, cookie state, account state, network route, and IP address. An estimated stream effect is therefore* not *an IP effect. Attributing it to the IP address alone would require a crossover factorial design that varies IP while holding the remaining factors fixed. We are explicit about this because the applied literature sometimes treats distinct IP addresses as independent sampling units, which the design does not license.*

Remark 2.2 (Why the collection day is the load-bearing facet). *The design treats the collection day as a facet in its own right because the platform appears to resample on a daily cycle rather than per request. Djorno et al. (2026) report that repeated identical requests are served from a cache for twenty-four hours and that a fresh sample is drawn when the cache resets at midnight UTC.*

Two consequences follow. First, repeated retrieval within a day cannot buy independent information about the sampling error, so the effective rate at which evidence accumulates is bounded at one fresh draw per day. Any procedure that reduces sampling variability by averaging repeated pulls — including the one recommended by Eichenauer et al. (2022) — therefore trades precision against calendar time, which is what makes it awkward for real-time work.

Second, the mechanism yields a prediction this framework can be judged by. If the daily cache holds exactly, the technical-replicate variance component in section 7 should be indistinguishable from zero, and the exact duplicates retained in section 6 are not a nuisance to be cleaned away but the direct observable signature of the cache. If that component is bounded away from zero, the daily-cache description is incomplete for the queries at hand. We state the prediction because it is cheap to test and because both outcomes are informative; we do not assume it, and no result in this paper depends on it holding.

2.2 Three failure modes of naive construction

Let X_t denote the latent relative search signal we wish to recover.

(F1) Compounding scale error. Under sequential stitching with estimated log-scale corrections $\hat{\ell}_j$, the calibrated value of chunk J carries the error $\sum_{j=2}^{J}(\hat{\ell}_j - \ell_j)$. Even if each join is unbiased, the variance of the accumulated error grows with the number of joins, so the oldest or newest part of a long series is systematically less reliable than the middle.

(F2) Dependence masquerading as replication. If N retrievals share a common retrieval-side disturbance, the variance of their mean does not fall as $1/N$. Reporting N as a sample size overstates precision. In the extreme case of exact duplicates, N retrievals contain the information of one.

(F3) Attenuation downstream. If the constructed index enters a regression as a covariate measured with classical additive error of variance σ_u^2, the OLS slope is attenuated by the reliability ratio $\lambda = \sigma_W^2/(\sigma_W^2 + \sigma_u^2)$. Measurement work that stops at the construction stage leaves this bias in place.

2.3 What the existing literature establishes

Each failure mode above has been documented separately, and the contribution here is to place them in one measurement chain rather than to discover them. Eichenauer et al. (2022) identify sampling noise and frequency inconsistency in GT and propose repeated draws as the remedy; Cebrián and Domenech (2024) show that the error in repeated pulls scales with a term's popularity; Medeiros and Pires (2021) demonstrate that a single realization can flip a forecasting conclusion by chance. On construction, Bleher and Dimpfl (2022) knit multi-annual high-frequency series from overlapping windows, and West (2020) calibrates across queries with an anchor bank that also handles rounding and censoring at low volume. Hölzl et al. (2025) survey how GT is misused in the social sciences, including the practice of reading a constructed index as if it were a count.

Two strands describe the retrieval mechanism itself. Neumann et al. (2023), working with Google Health Trends, document the sampling mechanism and recommend aggregating related phrases with the Boolean `OR` operator to lift series above the privacy threshold. Most directly related to the present paper, Djorno et al. (2026) treat missing values, sampling variability, noise, and trend as a single preprocessing problem, and report that repeated requests are served from a cache that resets at midnight UTC, so that a fresh sample is drawn only once per day. Their remedy is hierarchical clustering, smoothing splines, and detrending, validated by the accuracy of downstream influenza forecasts.

On empirical comparison this paper is deliberately narrow. Only sequential stitching is compared directly (section 5.4), and the result is that the mechanism holds while the accuracy difference does not reach significance. West (2020)'s anchor bank needs several queries to operate and Djorno et al. (2026) evaluate against downstream forecast accuracy, neither of which the single-series, latent-truth simulation here can accommodate. We therefore make no claim of accuracy superiority over those methods. What is established is a large improvement over downloading once, and a set of diagnostics the alternatives do not provide.

What that work leaves open is the step from a smoothed curve to an input for inference. None of it fixes the retrieval design in advance as a crossed experiment, decomposes error across collection day and stream, downweights dependent pulls by an estimated residual covariance, or carries the resulting uncertainty into the downstream regression. The distinction from Djorno et al. (2026) is the sharpest available statement of what this paper adds: that work seeks a cleaner curve and is evaluated by whether forecasts improve, whereas the target here is the measurement uncertainty itself, which is estimated, reported, and then propagated. A denoised series with no attached uncertainty is still consumed downstream as if it were measured without error. Once it is not, the downstream problem is the generated-regressor problem of Pagan (1984): a covariate that is itself an estimate makes the conventional standard error wrong even when the point estimate is serviceable. section 12 therefore treats the constructed series as generated rather than observed, in the errors-in-variables tradition of Carroll et al. (2006) and Cook and Stefanski (1994). Those are the gaps this paper addresses, and section 14 states precisely which of them come with proofs and which come with diagnostics only.

Remark 2.3 (Why the website version still needs chunking). *Google documents that Trends normalizes a sample of search activity by time and region and scales it to* 0–100*, with low volumes reported as zero (Google Trends Help, n.d.). The 2025 API alpha offers a consistent scale across requests, but its historical range is a rolling window of roughly 1,800 days and its values remain relative rather than absolute (Google Search Central, 2025). A study spanning 2010–2026 therefore still reconstructs its daily history from multiple shorter windows, which is what makes the calibration problem in section 5 unavoidable rather than optional.*

3 Estimands and identification

3.1 Two estimands

Definition 3.1 (Retrieval-mixture expectation). *Fix the query specification* Q *and a retrieval design that places known probability* π_h *on design cell* $h \in \mathcal{H}$*. The* retrieval-mixture expectation *is*

$$\theta_t^\pi \;=\; \sum_{h\in\mathcal{H}} \pi_h \, \mathbb{E}[Y_t \mid Q,\, h]\,, \tag{1}$$

where the inner expectation is taken over the GT system's own randomness conditional on the cell.

Definition 3.2 (Latent common-scale signal). *Let* Z_{it} *denote the calibrated value of pull* i *at date* t *after within-pull scale alignment and baseline normalization. The* latent common-scale signal X_t *is defined by the measurement model*

$$\boldsymbol{Z}_t \;=\; \mathbf{1}X_t + \boldsymbol{e}_t, \qquad \mathbb{E}[\boldsymbol{e}_t] = 0, \qquad \text{Var}(\boldsymbol{e}_t) = \boldsymbol{\Sigma}. \tag{2}$$

Definition 3.1 requires no model of the retrieval error and is estimable by stratified averaging. Definition 3.2 requires the assumptions below but supports variance-optimal estimation.

Remark 3.1 (The bridge from θ_t to X_t). *Definition 3.1 and definition 3.2 are not the same target, and two conditions connect them. First, proportionality:* $\theta_t = cX_t$ *for some* $c > 0$ *that does not depend on* t*, so that the expected GT response under a locked protocol tracks the latent relative signal. This rules out a systematic algorithmic bias that varies over time, and repeated measurement cannot test it. Second, negligible calibration error:* $Z_{it} = Y_{it}\,\text{exp}(-\hat{\ell}_{j(i)})$ *is a nonlinear transform of an* estimate*, and* $\mathbb{E}[Y\,\text{exp}(-\hat{\ell})] \neq a^{-1}\mathbb{E}[Y]$ *in general because* $\text{exp}(\cdot)$ *is convex. The first-order correction in section 5 leaves* $\mathbb{E}[\text{exp}(-\hat{\ell}_j)] = a_j^{-1}\{1 + O(n^{-1})\}$ *with* n *the number of usable overlap days, so* $\mathbb{E}[\boldsymbol{e}_t] = 0$ *in* (2) *is an* $O(n^{-1})$ *approximation rather than an exact statement. The unbiasedness and minimum-variance results below should be read under both conditions: the first needs external validation, the second improves as the overlap design is densified.*

Remark 3.2 (Why no additive pull bias is removed by default). *Version 1.0.0 subtracted* $b_i = T^{-1}\sum_t (Z_{it} - m_t)$ *from each pull, with* m_t *the daily median. Baseline normalization has already set* $T^{-1}\sum_t Z_{it} = 100$ *for every pull, so when the baseline spans the full history — the default that* `QuerySpec` *back-fills — that centring term reduces to* $b_i = 100 - T^{-1}\sum_t m_t$*, the* same constant *for every* i*. It removes no pull-specific difference; it shifts the entire consensus by the gap between the mean and the median, and it contradicts* $\mathbb{E}[\boldsymbol{e}_t] = 0$ *in* (2)*. In the controlled simulation it moved mean bias from machine precision to* -0.1464 *and degraded RMSE in all twenty replications. Version 1.1.0 therefore defaults* `center_pull_bias` *to off and reports* `pull_bias_cross_pull_sd`*, which is near zero exactly when the term car-*

ries no pull-specific information. Enable it only with baseline rescaling switched off and a genuine additive offset expected.

3.2 Assumptions

Assumption 3.1 (Protocol invariance). *Every retrieval in a batch uses the identical query specification, historical endpoints, and chunk manifest, and the configuration hash is unchanged across the batch.*

Assumption 3.2 (Value-independent inclusion). *Whether a retrieval is included in the analysis does not depend on the values it returned. Retrievals may be excluded for documented operational reasons recorded before inspection of the values.*

Assumption 3.3 (Positive-overlap multiplicative model). *For a date t present in chunks j and k of the same pull, with both raw values at or above a prespecified threshold $c_{\min}$,*

$$Y_{jt} = a_j X_t \exp(\epsilon_{jt}), \qquad a_j > 0, \tag{3}$$

so that with $\ell_j = \log a_j$ and $\nu_{jkt} = \epsilon_{jt} - \epsilon_{kt}$,

$$r_{jkt} \;=\; \log Y_{jt} - \log Y_{kt} \;=\; \ell_j - \ell_k + \nu_{jkt}. \tag{4}$$

Assumption 3.4 (Connected overlap graph). *The graph $G = (V, E)$ with V the set of chunks and E the set of chunk pairs having at least $m_{\min}$ usable overlapping days is connected.*

Assumption 3.5 (Conditionally centred edge error). $\mathbb{E}[\nu_{jkt} \mid (j,k) \in E] = 0$ *for every edge, and the edge-level estimator is applied to a set of overlap days selected without reference to the realized differences.*

Assumption 3.6 (Common signal with centred retrieval error). *Model* (2) *holds with Σ finite and positive definite on the retained pulls.*

Remark 3.3 (Where each assumption can fail). *Assumption 3.3 fails near zero, because rounding makes the log ratio explode; this is why only cells at or above $c_{\min}$ contribute to edges, while zeros are never imputed to positive values. One consequence of the aggregation rule must be stated plainly: when a date is covered by several chunks and only some of them return zero, the weighted average is positive, so that date is no longer zero in the final series. Zeros survive only where every covering chunk is zero. That is a corollary of averaging rather than an imputation decision, but it changes what detection reliability is measured on, so*

`CalibrationResult.diagnostics` *reports* `n_dates_with_zero_chunk` *and* `n_dates_zero_masked_by_aggregation` *for audit. Assumption 3.4 fails if the chunk design leaves a gap; the implementation raises an error rather than silently returning a partially identified series. Assumption 3.6 fails if a subset of pulls shares a retrieval-side bias that is not shared by the rest; this is precisely what the near-duplicate diagnostics in section 6 are designed to expose.*

4 Stage 0: preregistered acquisition design

The design fixes, before collection begins: the query specification; the historical endpoints; the chunk manifest (window length, step, minimum overlap); the set of collection-day offsets; the set of streams; the number of technical replicates; the time slots; and the acceptance thresholds. Stream order is rotated across collection days and chunk order is rotated across pulls, so that order effects are balanced rather than confounded with the facets of interest.

The entire configuration is serialized canonically and hashed:

$$h \;=\; \text{SHA256}\big\{\text{canonical}(Q, \mathcal{D}, \mathcal{C}, \mathcal{T})\big\}, \tag{5}$$

where $\mathcal{C}$ collects the calibration settings and $\mathcal{T}$ the decision thresholds. Every raw observation may store h. Any change to a locked setting changes h, so configuration drift between the design and the analysis is detectable rather than assumed away.

Pilot separation. An initial monitoring window may be used to check for gaps, query drift, exact duplicates, and connectivity. If that inspection leads to any change in thresholds, stitching rules, query, code, or decision tree, the monitored data must be relabelled as pilot and the confirmatory batch restarted. Otherwise the same data are used both to design and to validate the acceptance rule, producing data-dependent acceptance bias. For the same reason, all GT processing rules must be locked before any return, volatility, spread, or VaR result is examined; selecting the construction that gives the most significant financial result converts measurement validation into outcome-guided model selection.

5 Stage 1: global overlap-graph calibration

5.1 Robust edge estimation

For each chunk pair (j, k) with at least m_{min} usable overlap days, the observed log ratios $\{r_{jkt}\}$ are summarized by a Huber M-estimator of location (Huber, 1964), which solves

$$\sum_t \psi_c\left(\frac{r_{jkt} - \mu}{\hat{\sigma}}\right) = 0, \qquad \psi_c(u) = \max\{-c, \min(c, u)\}, \tag{6}$$

with $c = 1.345$ and $\hat{\sigma}$ a robust scale. The estimator returns the location $\hat{r}_{jk}$, a robust scale, and a variance $\hat{v}_{jk}$ that becomes the edge precision. Robustness matters here because a small number of overlap days can be contaminated by revisions or by the rounding boundary, and a sample mean of log ratios is not resistant to those points.

5.2 Graph weighted least squares

Collect the edges into a design matrix B with rows indexed by edges: row e for edge (j, k) has $+1$ in column j, -1 in column k, and zero elsewhere. Because only differences are identified, one chunk is fixed as the reference by setting $\ell_{\text{ref}} = 0$; the implementation selects the highest-degree node, which minimizes the leverage of any single poorly estimated edge. With $\boldsymbol{r}$ the vector of edge estimates and $\Omega = \text{diag}(\hat{v}_e)$,

$$\hat{\boldsymbol{\ell}} \;=\; \arg\min_{\boldsymbol{\ell}} \,(\boldsymbol{r} - B\boldsymbol{\ell})^\top \Omega^{-1} (\boldsymbol{r} - B\boldsymbol{\ell}) \;=\; (B^\top \Omega^{-1} B)^{-1} B^\top \Omega^{-1} \boldsymbol{r}. \tag{7}$$

Proposition 5.1 (Identification). *Under assumption 3.4, B has rank $|V| - 1$ after the reference normalization, so $\hat{\boldsymbol{\ell}}$ in (7) is uniquely defined.*

Proof. B is the incidence matrix of G. For a graph with $|V|$ nodes, the rank of the incidence matrix equals $|V|$ minus the number of connected components. Connectivity gives one component, hence rank $|V| - 1$. Deleting the reference column removes exactly the one-dimensional null space spanned by 1, leaving a full-column-rank matrix and a positive definite normal matrix $B^\top \Omega^{-1} B$. □

Proposition 5.2 (Conditional unbiasedness and efficiency). *Under assumptions 3.3 to 3.5, and treating Ω as known, $\hat{\boldsymbol{\ell}}$ is unbiased for $\boldsymbol{\ell}$ and is the minimum-variance* *estimator within the class of linear unbiased estimators of* $\boldsymbol{\ell}$ *based on the edge estimates.*

Proof. Write $\boldsymbol{r} = B\boldsymbol{\ell} + \boldsymbol{\nu}$ with $\mathbb{E}[\boldsymbol{\nu}] = 0$ by assumption 3.5 and $\mathrm{Var}(\boldsymbol{\nu}) = \Omega$. This is a linear model with full-column-rank design after normalization, so the Gauss–Markov theorem applies directly to the weighted least-squares estimator (7). □

That proposition says only that graph WLS is best among linear estimators built on the edge estimates. It does not answer the question the design actually turns on — *how much better than sequential stitching* — and failure mode F1 was until now an argument in words. The next proposition makes it an identity, and one that depends on the chunk design alone rather than on how noisy the retrievals turn out to be.

Proposition 5.3 (Calibration variance is an effective resistance). *Read the overlap graph as an electrical network in which edge* e *carries a* conductance *equal to its precision,* $w_e = 1/\,\mathrm{Var}(\hat{\delta}_e)$*. Then for any chunk* j *and reference* r*,*

$$\mathrm{Var}(\hat{\ell}_j - \hat{\ell}_r \mid B) = R_{\text{eff}}(j, r), \tag{8}$$

the effective resistance between j *and* r *under the weighted Laplacian* $L = B'WB$*. Two consequences follow.*

(i) Sequential stitching is the same estimator restricted to a spanning path, whose effective resistance is the plain sum *of the resistances it walks and therefore grows with the number of joins. That is F1, stated exactly rather than by analogy.*

(ii) (Rayleigh monotonicity) Adding an edge cannot raise the effective resistance between any pair of nodes. Hence

$$R_{\text{eff}}^{\textit{graph}}(j, r) \;\le\; R_{\text{eff}}^{\textit{path}}(j, r) \qquad \textit{for every } j, \tag{9}$$

so global graph calibration is never worse than any sequential stitch in calibration variance, and the margin can be computed from the design before *any data is collected.*

Proof. Grounding at r and deleting its row and column leaves the grounded Laplacian $B'_{-r}WB_{-r}$, whose inverse is the estimator covariance in (7). In network theory the jth diagonal entry of a grounded Laplacian inverse is the effective resistance from j to ground, which is (8). Consequence (i) is resistances in series. Consequence (ii) is Rayleigh's monotonicity theorem: an added edge is an added parallel path, which cannot increase the resistance between any two points. □

Remark 5.1 (F1 becomes a quantity the design stage can compute). *Since* $\mathrm{Var}(\hat{\delta}_e) \approx \sigma^2/n_e$ *for an overlap of* n_e *days with per-day log-ratio dispersion* σ*, the* σ^2 *cancels in the ratio of path to graph: the variance reduction is a function of the chunk design alone. Converting to a terminal scale error at the* $\sigma = 0.0188$ *measured on real data:*

Table 1: Calibration error implied by a chunk design through (8), under $\mathrm{Var}(\hat{\delta}_e) = \sigma^2/n_e$ at the median dispersion measured on one year of one keyword. Only the first row can be checked against data; the rest are extrapolations — see remark 5.2.

Design (span, window/step)	Chunks	Reduction	Sequential	Graph
Measured, from the estimated edge variances				
1 year, 180/30	8	9.13×	—	—
Extrapolated, from the idealized edge model				
1 year, 180/30	8	$10.5\times$	0.40%	0.12%
5 years, 180/30	56	$18.2\times$	1.15%	0.27%
15 years, 180/30	178	$20.1\times$	2.06%	0.46%
15 years, 180/15 (recommended)	355	$140.5\times$	2.79%	0.23%

Remark 5.2 (What in this table is measured and what is not). *The identity* (8) *and the ordering* (9) *are theorems and need no data. The* numbers *in the table are a different matter, and two of their assumptions are contradicted by the same year of data they are calibrated on. Both contradictions run in the direction that flatters the conclusion, so they are stated rather than left for a reader to find.*

The idealized edge model overstates the reduction by about 15%. On the one design where both can be computed, the estimated edge variances give $9.13\times$ *against the model's* $10.5\times$*: five of the 26 overlaps are degenerate and take the variance floor rather than* σ^2/n_e*, and the Huber weights leave* $n_{\text{eff}} < n_e$*. The extrapolated rows have no such check.*

σ *is not a constant. Across the 21 informative overlaps it ranges over a factor of* 1.75 within a single year *and correlates* -0.760 *(*$p = 0.0001$*) with the value level of the pair, which is what the rounding floor* $1/(\sqrt{12}\,v)$ *predicts. A fifteen-year series spans far wider variation in search volume than one year of it does, so the low-volume eras carry a larger* σ *and the absolute percentages in the last three rows are understated. The* ratio *is more robust, because* σ *largely cancels between path and graph, but it does not cancel exactly once* σ *varies across edges.*

The reach is also worth stating plainly: one year, one keyword, one pull, extrapolated fifteen-fold. The claim that F1 is material at long horizons is a prediction of proposition 5.3, not a measurement. Testing it needs a long-horizon collection, which *section 4 specifies and this manuscript has not carried out.*

From version 1.4.0 `CalibrationResult.diagnostics` *reports* `max_log_scale_variance`, `sequential_max_log_scale_variance`, *and their ratio* `calibration_variance_reduction`*.*

Remark 5.3 (Why this dominates sequential stitching). *Sequential stitching is the special case in which only the $J-1$ consecutive edges are used and each is applied in turn. Proposition 5.2 uses every edge, including the non-consecutive overlaps that a long window and a short step generate, and weights each by its estimated precision. It also makes error accumulation impossible: no estimate is defined as a function of a previously calibrated estimate. In addition, the residual $\boldsymbol{r} - B\hat{\boldsymbol{\ell}}$ has a direct interpretation as cycle inconsistency, which is reported as a diagnostic and has no analogue in the sequential procedure.*

5.3 Reconstruction and normalization

Calibrated observations are $\tilde{Y}_{jt} = Y_{jt}\exp(-\hat{\ell}_j)$, optionally with a lognormal bias correction. Overlapping dates are aggregated by inverse-variance weighting. The resulting series is normalized so that its mean over the prespecified baseline window equals 100. This normalization defines the scale of the estimand; it is not an estimate of absolute search volume.

5.4 A direct comparison with sequential stitching

Failure mode F1 argued that sequential stitching accumulates scale error along the chain and that solving the graph avoids it. Until now that was an argument, not a measurement. `src/racergt/baselines.py` implements sequential stitching in the style of Bleher and Dimpfl's knitting, reusing the *same* Huber edge estimator, the same $c_{\min}$ gate, and the same baseline normalization; the only difference is that it walks a spanning path instead of solving the whole graph, so any gap is attributable to that choice alone. `examples/run_calibration_comparison.py` runs both.

Table 2: Global graph calibration against sequential stitching, 20 seeds by 2 pulls per cell. Advantage is sequential minus graph in RMSE, so positive favours the graph.

Design	Joins	Graph	Sequential	Advantage	MCSE	p	Graph wins
60-day step	5	7.4827	7.4677	-0.0150	0.0053	0.008	13/40
30-day step	9	7.3026	7.2967	-0.0059	0.0049	0.234	15/40
15-day step	17	7.2085	7.2125	$+0.0040$	0.0072	0.576	15/40
15-day step, low noise	17	1.9245	1.9356	$+0.0111$	0.0066	0.098	21/40

The mechanism holds; its effect is second order. The accumulation F1 describes is real and its magnitude is clear: under the 15-day design the cross-replication standard deviation of the recovered log scale rises from 0.0259 early in the chain to 0.1228 at the end, a factor of 4.75, while the graph's ratio is 1.12. Error does accumulate along a spanning path, and solving jointly does prevent it.

It does not translate into a detectable advantage in reconstruction error. The direction matches F1 — going from 5 joins to 17 moves the advantage from -0.0150 to $+0.0040$, and cutting retrieval noise from 0.06 to 0.01, so that calibration error carries a larger share of the total, raises it further to $+0.0111$ — but even in that setting, purpose-built to favour F1, $p = 0.098$ with 21 wins out of 40. At the shortest chain the graph is significantly *worse* ($p = 0.008$), because its reference chunk is chosen by highest degree rather than fixed at the first chunk, and because it additionally applies the lognormal correction.

Remark 5.4 (Three mechanisms, one conclusion). *This is the third result of the same shape. Covariance-adjusted weighting correctly identifies and down-weights dependent pulls (weights correlate* -0.562 *with shared residual dependence) without improving RMSE. The prediction that heterogeneous dependence would change that was tested and failed. Global graph calibration does prevent error accumulating along the chain (*1.12 *against* 4.75*) and likewise does not improve RMSE.*

The common structure is that each mechanism works as designed and can be shown to work, while its effect on point accuracy *is second order and swamped by retrieval noise. For a user that conclusion matters more than any single number: the value of this framework is not that it produces a more accurate series, but that it produces a series whose quality can be judged. Effective pull counts, dependence structure, cycle-consistency statistics, per-day conditional standard errors, and propagatable measurement uncertainty are information that RMSE neither expresses nor replaces.*

5.5 Calibration on real Google Trends data

All of the evidence above comes from the package's own simulator, which draws from the same proportionality model the estimator assumes. That is circular: it can show that the estimator solves the problem it was handed, never that the problem is the right one. This section runs the calibration stage on real downloads — keyword `bitcoin`, United States, daily, calendar year 2024, eight chunks on 180-day windows with a 30-day step, 1,440 observations. The converter is `examples/ingest_trends_csv.py` and the analysis is `examples/run_real_data_calibration.py`.

What this data cannot test, stated first. It is a *single pull*: one collection day, one stream, one replicate. Every cross-pull mechanism is therefore *entirely* untested here — covariance-adjusted consensus weighting, exact and near-duplicate diagnosis, the three-facet variance components, effective pull counts, and the RMSE comparison of section 5.4 all require repeated collection across days and environments. This section does not touch them and claims nothing about them. Testing the full design requires the month-long collection schedule of section 4. Real data also carries no latent truth, so reconstruction error cannot be computed and neither method can be called more accurate.

5.5.1 The normalization creates zero-information overlaps

The first finding is one the simulator cannot structurally produce. Trends divides each chunk by the maximum inside *that request's own window*. When two windows contain the same peak day they are divided by the same number and, after rounding, are *identical cell for cell* wherever they overlap. Here C0001–C0003 all peak on 2024-03-05, C0004–C0005 on 2024-08-05, and C0007–C0008 on 2024-12-05, so 5 of the 26 overlap edges have a log ratio that is identically zero with no dispersion: the eight chunks form only four normalization groups.

Such an edge carries no scale information, yet it still enters the weighted least-squares fit at whatever weight `edge_variance_floor` and `max_edge_weight` permit — numerical safety constants, not a statistical argument. `simulate_racergt_data` multiplies every chunk by continuous lognormal noise *before* taking the window maximum, and at its default settings the chance that two chunks round to identical integers throughout an overlap is negligible: zero degenerate edges in 520 simulated, against 5 in 26 real ones. (Setting `chunk_noise_sd` to 0 would let the simulator produce them, but that is neither its default nor the setting under which the earlier evidence was generated.) Version 1.4.0 therefore adds `n_zero_dispersion_edges`, `n_informative_edges` and `n_scale_groups` to `CalibrationResult.diagnostics`. They report only and change no estimate; what to do about a degenerate edge is the analyst's judgment, not the package's.

The rate depends on the series' peak structure and cannot be extrapolated from one keyword. The mechanism is deductive — a shared window maximum implies an identical overlap — but the proportion is an empirical question.

5.5.2 The proportionality assumption survives

The whole calibration stage rests on overlapping chunks being proportional up to a constant. The overlaps run to 150 days, so the assumption is heavily over-identified and can be checked directly.

Table 3: Calibration diagnostics on real data. Bitcoin, United States, daily, 2024; eight chunks, one pull.

Diagnostic	Value
Overlap graph connected	yes
Edges (a spanning path uses 7)	26
Independent cycles	19
Testable triangles	45
Normal matrix condition number	10.82
Weighted edge RMSE	0.001379
Zero-dispersion edges	5 / 26
Normalization groups	4 (among 8 chunks)
Median triangle closure error	0.001783 log points
Maximum triangle closure error	0.011769 log points (1.184% scale error)
Overlaps with drift $p < 0.05$	1 / 21 (1.05 expected by chance)
Largest residual mis-scaling	-0.032% *(no power; see remark 5.5)*

Triangle closure is one of the framework's central advantages over sequential stitching, and stitching cannot supply it at all: the model forces $d_{ij} + d_{jk} = d_{ik}$ for any three mutually overlapping chunks, while a spanning path uses only the edges it walks and can neither see nor report a violation. Across 45 triangles the median closure error is 0.0018 log points and the largest is 0.0118, a 1.18% scale error. Only 1 of 21 informative overlaps shows drift at $p < 0.05$, against 1.05 expected by chance.

Remark 5.5 (What these diagnostics can and cannot detect). *A diagnostic that passes is worth nothing until it is shown capable of failing. Injecting a known* *perturbation into one chunk and re-running gives each test a minimum detectable effect.*

Table 4: Response of each diagnostic to a known perturbation of chunk C0005. Baseline maximum closure error is 0.0118; baseline drift count is 1 of 21.

Perturbation	Max closure error	Drift $p < 0.05$	Per-chunk deviation
Uniform rescale, any size	0.0118 (none)	1 (none)	none
Trend, 2% per 100 days	0.0232	7 of 22	none
Nonlinear, 2%	0.0118 (none)	5 of 22	none
Nonlinear, 5%	0.0212	5 of 22	none

The per-chunk deviation test has exactly zero power and is reported only to say so. Multiplying a chunk by a constant is absorbed entirely into $\hat{\ell}_j$, leaving $Y_{jt}\,\exp(-\hat{\ell}_j)$ unchanged, so injected errors of 2% and 20% return the same number to the digit. That is not a weakness of the estimator — a uniform rescale is the parameter, not a violation — but it means the -0.032% in table 3 is evidence of nothing.

Cycle closure and the drift test do have power, and against violations that a scale cannot absorb. The drift test is the more sensitive of the two, catching a 2% nonlinear distortion that closure misses. Reading the baseline against the table: the observed maximum closure error is half what a 2%-per-100-days trend would produce, so violations of that size or larger are excluded, which is the strongest statement this data supports. It is not the same as the model being correct.

5.5.3 The simulator overstates chunk noise by about 4.5 times

This data cannot test F1, but it decisively tests something else: whether the simulator describes reality. `SimulationSettings.chunk_noise_sd` defaults to 0.03, multiplying every chunk on every day by an independent lognormal disturbance before the maximum is taken. The real overlaps have a median log-ratio dispersion of 0.0188.

Subtracting a closed-form integer-rounding floor returns 0, but that floor assumes the two chunks round *independently*. They do not: they round the same numbers differing only by a scale factor, so their rounding errors are correlated, the closed form overstates itself, and the recovered noise is biased towards zero — the direction that flatters this conclusion. It is therefore not used. A parametric bootstrap replaces the assumption with the mechanism: take the calibrated daily series, multiply by chunk noise at a stated level, divide by the window maximum, round, and measure dispersion exactly as on the real chunks.

Table 5: Sweeping `chunk_noise_sd` against the observed overlap dispersion of 0.018814. Two hundred replications per level.

σ_{chunk}	Median overlap dispersion	Ratio to observed
0 (rounding only)	0.017757	0.94
0.005	0.017644	0.94
0.0066 (matched)	0.018814	1.00
0.010	0.021892	1.16
0.020	0.033624	1.79
0.030 (simulator default)	0.046424	2.47

Integer rounding alone accounts for 94% of the observed dispersion, and the matched value is $\sigma_{\text{chunk}} \approx 0.0066$, about 4.5 times smaller than the simulator's default. Below about 0.005 the curve is flat within Monte Carlo error — rounding dominates and the parameter is not identified there — but 0.03 produces 2.47 times the observed dispersion and is firmly excluded.

The sweep itself is checked the same way the diagnostics are in remark 5.5, by giving it something to find. Injecting a known disturbance into the real chunks, renormalizing the way Trends does, and rerunning recovers 0.0067, 0.0099, 0.0234 and 0.0312 at injected levels of 0, 0.01, 0.02 and 0.03. The last is what matters: had the simulator's default been the truth, the procedure would have returned about 0.03 rather than the small value it returns on untouched data, so the gap is not the estimator failing.

This *strengthens* rather than undermines section 5.4. That comparison was run in a regime where calibration noise was several times too large, and even in that F1-friendly regime the global graph secured no detectable accuracy advantage. Rerunning it at realistic noise would only shrink the gap.

5.5.4 Why one year cannot see F1, and when F1 starts to matter

The two methods are nearly indistinguishable on real data: correlation 1.000000, median daily relative difference 0.00026, maximum 0.00123, and a largest absolute difference of 0.195 points on the 0–100 index scale.

Remark 5.6 (A methodological error worth naming). *Testing F1 by asking whether the gap between the two point estimates grows along the chain is the* wrong estimator. *F1 is a claim about a variance, and a single pull supplies one realization; a variance cannot be estimated from $n = 1$, so that test has no power by construction, at any sample size. Proposition 5.3 removes the need to sample at all: the variance* *ratio is an effective-resistance ratio, a deterministic function of the design.*

Applying proposition 5.3 to this design, the graph's largest $\mathrm{Var}(\hat{\ell}_j)$ is 1.01×10^{-6} against 9.26×10^{-6} for the spanning path, a reduction of 9.13 times, and the reduction grows monotonically with position along the chain ($2.1\times$ to $9.1\times$). F1's mechanism is operating, and its size follows from the design without any test.

In absolute terms, however, the terminal error is 0.40% for sequential stitching and 0.12% for the graph. A large ratio and a small magnitude are not in tension, and together they explain both why one year sees nothing and why the three earlier simulation experiments returned nulls.

Remark 5.7 (F1 is predicted to be material at the recommended protocol). *Table 1 extrapolates the error out along the historical span. At 15 years with 180-day windows and a 15-day step — the specification recommended here — sequential stitching would carry a terminal scale error near* 2.79% *against* 0.23% *for the graph. On an attention index spanning fifteen years, a level error of that size at one end is a slow drift hard to separate from a trend: exactly the artificial low-frequency drift the Google Trends literature warns about. Remark 5.2 states what those two numbers rest on; they are a prediction of proposition 5.3 and have not been measured.*

The correct statement about F1 is therefore that the mechanism is real and provable (proposition 5.3), that it is predicted *to be material at the recommended scale, and that the one year measured here cannot see it because the design's absolute error ceiling is* 0.40%*. The nulls in section 5.4 and section 13 are not evidence against F1; they are what proposition 5.3 predicts, since on short chains the absolute difference between effective resistances must be small.*

This revises remark 5.4. Its reading was that each mechanism works while its effect is second order. Proposition 5.3 shows that for calibration the effect is not second order but scale-dependent *— it grows with the number of joins, and every experiment so far ran at the end of the range where it is smallest. Beyond the diagnostics in table 3 that a spanning path structurally cannot supply, the reason to prefer global calibration is the unconditional guarantee in* (9)*: it is never worse.*

6 Stage 2: duplicate and dependence diagnostics

Two retrievals may be numerically identical, or nearly so. Both cases violate the implicit assumption that repeated pulls contribute independent information, but they require different treatment.

Exact duplicates. Vectors are hashed after rounding to a fixed number of decimals. Exact groups are retained in the audit archive—deleting them would destroy evidence about the retrieval process—and are collapsed analytically before the consensus step, with multiplicity recorded.

Near-duplicates. Raw correlation between two pulls is uninformative, because all pulls share the same strong latent signal and are therefore correlated at close to one by construction. The diagnostic instead removes the common signal first and examines the residual correlation, together with value-anchored measures: mean absolute deviation on the 0–100 scale, exact cell agreement, and the Jaccard index of the positive-value sets. Pairs that exceed the prespecified thresholds form the edges of a dependence graph.

Remark 6.1 (Components are diagnostic, not equivalence classes). *Connected components of the dependence graph are reported as connectivity sets. Membership in a component does* not *imply that every pair inside it exceeds the threshold, because connectivity is transitive while the pairwise rule is not. The acceptance rule uses the share of the batch occupied by the largest component, which is the quantity that actually matters for whether the batch is dominated by one dependence cluster.*

7 Stage 3: generalizability decomposition

Classical reliability coefficients assign all non-signal variation to a single undifferentiated error term. Generalizability Theory (Cronbach et al., 1972; Brennan, 2001) instead decomposes it across the facets of the measurement design, which is what a researcher needs in order to decide where to spend the next unit of collection effort.

For a fully crossed design over historical date t, collection day d, stream s, and replicate r,

$$Y_{tdsr} = \mu + T_t + D_d + S_s + (TD)_{td} + (TS)_{ts} + (DS)_{ds} + (TDS)_{tds} + E_{tdsr}. \tag{10}$$

Variance components are estimated by the method of moments from the balanced ANOVA sums of squares. The implementation validates full crossing and equal cell counts before estimating, and raises an error otherwise, rather than silently producing components from an unbalanced design.

Remark 7.1 (Confounding with a single replicate). *When $n_r = 1$, E cannot be separated from (TDS); the implementation retains the combined term and reports it as such. This is stated rather than hidden because it changes the interpretation of the residual component.*

The decomposition is run on three transformations, which answer three different questions:

- level, Y itself: do relative magnitudes reproduce?
- detection, $\mathbb{I}\{Y > 0\}$: does the zero/non-zero pattern reproduce? This is the binding constraint for low-volume terms.
- innovation, $\Delta \log(1 + Y)$: do daily *changes* reproduce? This is the relevant question when the index enters a model in differences, and it is systematically harder to satisfy than the level criterion.

The generalizability coefficient for relative decisions and the dependability coefficient for absolute decisions follow in the usual way, and the D-study evaluates them on a grid of hypothetical designs, so the marginal value of an additional collection day can be compared with that of an additional stream.

8 Stage 4: the design-based reference estimator

Let h index the prespecified collection-day $\times$ stream cells with target weights π_h summing to one, and let n_h be the number of completed pulls observed in cell h. The stratified estimator is

$$\hat{\theta}_t^{\pi} \;=\; \sum_h \pi_h \frac{1}{n_h} \sum_{i \in h} Z_{it}. \tag{11}$$

Proposition 8.1 (Design unbiasedness). *Under assumptions 3.1 and 3.2, and provided $n_h \geq 1$ for every cell with $\pi_h > 0$, $\hat{\theta}_t^{\pi}$ is unbiased for the retrieval-mixture expectation θ_t^{π} of definition 3.1.*

Proof. Conditional on cell h, the retrievals in that cell are draws from the GT system under a fixed specification, so $\mathbb{E}[n_h^{-1} \sum_{i \in h} Z_{it}] = \mathbb{E}[Y_t \mid Q, h]$ by assumption 3.2, which guarantees that inclusion carries no information about the values. Taking the π-weighted sum over cells and applying linearity gives (1). $\square$

Remark 8.1 (Dependence affects the variance, not the bias). *Proposition 8.1 does not require the retrievals to be independent. Dependence across pulls inflates* $\mathrm{Var}(\hat{\theta}_t^{\pi})$

but leaves the expectation unchanged. This is exactly why the estimator is retained as a transparent reference: its validity rests on the design, not on a covariance model that could itself be misspecified.

9 Stage 5: covariance-adjusted consensus

9.1 Minimum-variance weights

Under assumption 3.6, consider linear estimators $\hat{X}_t = \boldsymbol{w}^\top \boldsymbol{Z}_t$ with $\boldsymbol{w}^\top \mathbf{1} = 1$, which is the condition for unbiasedness.

Proposition 9.1 (BLUE weights). *Under assumption 3.6 with $\boldsymbol{\Sigma}$ known and positive definite,*

$$\boldsymbol{w}^* = \frac{\boldsymbol{\Sigma}^{-1}\mathbf{1}}{\mathbf{1}^\top \boldsymbol{\Sigma}^{-1}\mathbf{1}} \tag{12}$$

solves $\min_w \boldsymbol{w}^\top \boldsymbol{\Sigma} \boldsymbol{w}$ *subject to* $\boldsymbol{w}^\top \mathbf{1} = 1$, *and the resulting* $\hat{X}_t = \boldsymbol{w}^{*\top} \boldsymbol{Z}_t$ *has variance* $(\mathbf{1}^\top \boldsymbol{\Sigma}^{-1}\mathbf{1})^{-1}$, *the minimum over all linear unbiased estimators.*

Proof. The Lagrangian is $L = \boldsymbol{w}^\top \boldsymbol{\Sigma} \boldsymbol{w} - 2\lambda(\boldsymbol{w}^\top \mathbf{1} - 1)$ with first-order condition $\boldsymbol{\Sigma}\boldsymbol{w} = \lambda\mathbf{1}$, hence $\boldsymbol{w} = \lambda \boldsymbol{\Sigma}^{-1}\mathbf{1}$. Imposing $\boldsymbol{w}^\top \mathbf{1} = 1$ gives $\lambda = (\mathbf{1}^\top \boldsymbol{\Sigma}^{-1}\mathbf{1})^{-1}$ and (12). The objective is strictly convex because $\boldsymbol{\Sigma} \succ 0$, so the stationary point is the unique minimum. Substituting yields the stated variance. □

9.2 Stabilization in finite samples

Proposition 9.1 assumes $\boldsymbol{\Sigma}$ known. In practice the number of pulls is of the same order as, or smaller than, the number of dates used to estimate the covariance, so the sample covariance is ill-conditioned and its inverse is unstable. The implementation therefore:

1. collapses exact duplicates analytically, so that a repeated vector cannot inflate the apparent rank;
2. removes a pull-level offset, so that a constant retrieval bias is not mistaken for signal disagreement;
3. estimates $\boldsymbol{\Sigma}$ by Ledoit–Wolf shrinkage (Ledoit and Wolf, 2004) toward a well-conditioned target;
4. imposes $w_i \geq 0$ and $w_i \leq \bar{w}$ while retaining $\sum_i w_i = 1$.

Remark 9.1 (What the constraints cost). *The nonnegativity and cap constraints are stabilization devices, not improvements on proposition 9.1. The strict BLUE property belongs to the unconstrained solution. The constrained solution is a minimum-variance convex combination: it gives up the theoretical optimum in exchange for refusing the large offsetting positive and negative weights that an ill-conditioned* $\hat{\boldsymbol{\Sigma}}^{-1}$ *would otherwise produce. We regard this as the right trade for a measurement instrument that must behave predictably on data the analyst has not yet seen.*

9.3 Effective number of pulls

Because dependence is the central concern, the raw count of pulls is not reported as a sample size. We report instead the Kish effective count (Kish, 1965) $1/\sum_i w_i^2$ and a spectral effective count derived from the eigenvalues of the residual correlation matrix. Both fall below the nominal count when retrievals are dependent, and the acceptance rule in section 11 is stated in terms of the spectral count.

The spectral count carries a known downward bias that should be read alongside it. Residuals are formed by subtracting the daily median *of the pulls themselves*, which induces negative correlation among them, so the residual correlation matrix is not the identity even when the underlying pulls are independent. On independently generated series the estimate averages 8.57 at $m = 9$ and 2.59 at $m = 3$, shortfalls of 4.8% and 13.7%. The direction is conservative — it understates information and therefore errs toward rejecting a batch — but it means the effective strictness of `min_spectral_effective_pulls` exceeds its nominal value, most visibly in small designs.

10 Stage 6: temporal benchmarking

Long GT histories at daily frequency are not directly comparable to the weekly or monthly series that GT returns for the same query over the full window. Let $\boldsymbol{z}$ be the preliminary daily consensus, A the aggregation matrix mapping days to benchmark periods, and $\boldsymbol{b}$ the lower-frequency benchmark. The soft benchmark solves

$$\min_{\boldsymbol{x}} \, (\boldsymbol{x} - \boldsymbol{z})^\top Q(\boldsymbol{x} - \boldsymbol{z}) \; + \; (A\boldsymbol{x} - \boldsymbol{b})^\top W(A\boldsymbol{x} - \boldsymbol{b}), \tag{13}$$

where $Q = (\lambda_I + \lambda_R)I + \lambda_D D_1^\top D_1$ combines fidelity to the daily consensus with a *first*-difference smoothness penalty on the correction path, and W is the benchmark precision. The ridge λ_R is a separate configuration field whose only role is to keep Q positive definite when $\lambda_I = 0$, so the two identity terms are one coefficient mathematically. The default $\lambda_D/\lambda_I = 10$ is a strong smoothing prior: it pushes corrections toward low-frequency adjustments and can systematically damp daily shape near event days, which is where an attention index is most informative. Calibrate and lock that ratio in a pilot, and report sensitivity over $\lambda_D/\lambda_I \in \{1, 10, 100\}$. This is the quadratic-programming form of the movement-preservation idea familiar from Denton (1971) and Chow and Lin (1971); treating $\boldsymbol{b}$ as noisy rather than exact is appropriate here because the lower-frequency GT series is itself a sampled quantity.

Proposition 10.1 (Existence and uniqueness). *If $Q \succ 0$ and $W \succeq 0$, the objective in* (13) *is strictly convex and the solution is the unique $\boldsymbol{x}^* = (Q + A^\top W A)^{-1}(Q\boldsymbol{z} + A^\top W \boldsymbol{b})$.*

Proof. The Hessian is $2(Q + A^\top W A)$, which is positive definite because $Q \succ 0$ and $A^\top W A \succeq 0$. Setting the gradient to zero gives the stated normal equations, whose coefficient matrix is invertible. □

11 Stage 7: reliability and the acceptance decision tree

Three reliability questions are kept separate because they can have different answers for the same series: whether the zero/non-zero pattern reproduces (detection, summarized by Fleiss' κ, Fleiss, 1971), whether levels reproduce, and whether daily innovations reproduce.

The acceptance decision is a locked tree evaluated after estimation. Each rule compares an observed diagnostic against a prespecified threshold and returns pass, fail, or indeterminate. The batch status is

- FAIL if any mandatory rule fails;
- REVIEW if no mandatory rule fails but at least one is indeterminate;
- PASS otherwise.

The rules cover protocol integrity, overlap-graph connectivity, the number of numerically distinct pulls, the spectral effective pull count, the share of zero observations, level and innovation generalizability, detection agreement, concentration of the near-duplicate dependence graph, consensus convergence as the last pull is added, and consistency with the lower-frequency benchmark.

Remark 11.1 (Indeterminate is not a soft fail). *The detection rule is not identified when every observation falls in one detection category, because κ is undefined without variation. The implementation reports the rule, marks it indeterminate, and—crucially—makes it non-mandatory in exactly that case, so that a series with no zeros is not penalized for a coefficient that cannot exist. Conflating "cannot be computed" with "failed" would systematically reject high-volume terms.*

Remark 11.2 (What PASS does and does not mean). *PASS authorizes construction under the locked protocol. It does not establish that the keyword has construct validity for the economic concept of interest, it does not identify absolute search counts, and it licenses no causal interpretation.*

12 Stage 8: propagating measurement error downstream

Let $W_t = X_t + u_t$ be the constructed index, used as a regressor for an outcome Y_t with controls. Under classical additive measurement error, the moment-corrected estimator is

$$\hat{\beta} = \frac{\boldsymbol{W}^{\top} M \boldsymbol{Y}}{\boldsymbol{W}^{\top} M \boldsymbol{W} - \mathrm{tr}(M\Omega_u)}, \tag{14}$$

where M is the residual maker for the controls and Ω_u the measurement error covariance supplied by the construction stage. Standard errors use a HAC estimator (Newey and West, 1987). The implementation raises an error when the corrected denominator is non-positive, rather than returning an explosive or sign-flipped coefficient; a non-positive denominator means the estimated measurement error exceeds the observed regressor variation, which is a statement about the data, not a numerical detail to be smoothed over.

As an alternative that does not require the classical structure to hold exactly, SIMEX (Cook and Stefanski, 1994; Carroll et al., 2006) adds further known noise on a grid of λ values, records the resulting estimates, fits an extrapolation function, and evaluates it at $\lambda = -1$.

13 Controlled Monte Carlo evidence

The simulation generates a latent signal with realistic dynamics, applies collection-day and stream effects, adds correlated retrieval error, splits the history into independently normalized chunks, imposes rounding and zeros, and optionally injects exact duplicates. Because the latent truth is known by construction, estimators can be compared directly.

13.1 Matching the information sets

`PipelineResult.final_series` returns the benchmarked series whenever a benchmark is available. Comparing that series against cross-pull baselines that never received the benchmark does not compare four estimators; it compares three estimators against one estimator plus an extra block of low-frequency information, and in simulation that block is generated from the latent truth itself. Every estimator is therefore evaluated twice: on the calibrated pulls alone, and after applying the *same* weekly and monthly benchmark. The single-pull row reports mean performance over all pulls rather than one fixed pull, because a fixed column measures which pull happened to sort first, not what a researcher who downloads once should expect.

Table 6: Twenty controlled replications (seeds 1200–1219). MCSE is the Monte Carlo standard error of the mean RMSE. Lower RMSE is better.

Estimator	Benchmark	RMSE	MCSE	Corr.	Innov. corr.	Peak recall
Single pull	no	7.3519	0.0573	0.9688	0.8878	0.8912
Cross-pull median	no	4.1710	0.0819	0.9898	0.9556	0.9447
Simple mean	no	3.7253	0.0877	0.9919	0.9747	0.9526
RACER-GT	no	3.7421	0.0894	0.9918	0.9743	0.9526
Single pull	yes	6.9881	0.0491	0.9717	0.8882	0.8947
Cross-pull median	yes	3.9584	0.0695	0.9908	0.9558	0.9474
Simple mean	yes	3.5058	0.0754	0.9928	0.9748	0.9553
RACER-GT	yes	3.5226	0.0770	0.9927	0.9745	0.9526

Table 7: Paired comparisons on RMSE. A positive difference favours estimator A; $n = 20$ paired replications.

Estimator A	Estimator B	Difference	t	p	A wins
RACER-GT	Single pull	$+3.6098$	48.31	2.4×10^{-21}	20/20
RACER-GT	Cross-pull median	$+0.4289$	14.86	6.5×10^{-12}	20/20
RACER-GT	Simple mean	-0.0167	-1.49	0.152	7/20
RACER-GT (bm.)	Simple mean (bm.)	-0.0168	-1.65	0.116	7/20
RACER-GT (bm.)	RACER-GT (no bm.)	$+0.2194$	15.49	3.1×10^{-12}	20/20
Simple mean (bm.)	Simple mean (no bm.)	$+0.2195$	15.45	3.3×10^{-12}	20/20

Three conclusions follow, two of which support the framework and one of which does not. First, cross-pull aggregation buys a great deal: at a matched information set RACER-GT cuts RMSE by $1 - 3.7421/7.3519 = 49.1$% against a single pull, improving in all twenty replications, and by a further 10.3% against the cross-pull median. Second, covariance-adjusted weighting shows no detectable advantage over the simple mean, before benchmarking (-0.0167, $p = 0.152$) or after it (-0.0168, $p = 0.116$). Third, temporal benchmarking contributes about 0.22 on its own, and contributes it almost identically to RACER-GT and to the simple mean (0.2194 versus 0.2195): the gain belongs to the benchmark, not to the daily estimator it is paired with.

Remark 13.1 (A testable prediction). *The second conclusion is a negative result, and its most natural explanation is that dependence in this DGP is homogeneous: every pull shares its collection-day noise with fixed weight* 0.5*, so the residual covariance is close to equicorrelated, the minimum-variance weights of an equicorrelated matrix are equal weights, and GLS has nothing to exploit while still paying for estimating* Σ*.*

If that explanation holds, GLS should win once dependence strength differs *across pulls — when some pulls are served from one cache and others are not. That is a testable prediction, and section 13.2 tests it directly.*

13.2 A direct test of dependence heterogeneity

`examples/run_dependence_experiment.py` places a subset of pulls behind one shared cache disturbance. Membership cuts across collection days and streams, so the extra dependence *cannot* be absorbed by the design facets and is visible only in the residual covariance — which is what the GLS weights exist for. At `cache_cluster_fraction` $= 0$ the simulated data are byte-identical to those behind table 6, so the two sections are directly comparable.

Table 8: Four dependence scenarios, 20 replications each. Advantage is simple mean minus GLS in RMSE, so positive favours GLS.

Scenario	GLS	Simple mean	Median	Advantage	MCSE	p	GLS wir
Homogeneous (baseline)	3.7421	3.7253	4.1710	-0.0167	0.0112	0.152	7/20
1/3 of pulls, moderate	3.7658	3.6801	4.1133	-0.0857	0.0155	< 0.0001	2/20
1/3 of pulls, strong	4.1684	4.1980	4.6486	$+0.0296$	0.0439	0.508	10/20
1/2 of pulls, strong	5.1633	4.8802	5.5593	-0.2831	0.0591	0.0001	2/20

The prediction fails. GLS wins no scenario significantly. The closest case (one third of pulls, strong dependence) is $+0.0296$ with $p = 0.51$ and exactly 10 wins out of

20, which is a tie; in two scenarios GLS is *significantly worse*. Adding heterogeneity does not reverse the ranking.

Two diagnostics rule out the obvious misreadings of that result.

First, the constraints are not the cause. Relaxing them under the strong one-third scenario (10 replications), the constrained default, the uncapped variant, and the fully unrestricted solution all attain a mean RMSE of 4.1752 — identical to the digit, because the unrestricted solution already lies inside the feasible set. An empirical covariance gives 4.1785 and a diagonal one 4.2345, against 4.2415 for the simple mean over the same seeds. The unrestricted estimator, which is BLUE in theory, does not win either.

Second, the mechanism works. The correlation between a pull's mean residual correlation and the weight it receives is -0.562, negative in all ten replications: GLS does find the pulls that share dependence and does move weight away from them. Weight dispersion in table 8 rises monotonically with heterogeneity ($0.0220 \to 0.0409$) and the spectral effective pull count falls monotonically ($4.93 \to 2.95$); both correctly register the added dependence.

What remains is estimation error. Σ must be estimated from 9 pulls over 366 historical dates, and the perturbation that its error induces in the weights exceeds the efficiency that exploiting the true dependence structure can buy. This matches a long-standing finding in the portfolio literature, where sample-covariance minimum-variance portfolios fail to beat equal weighting in small samples; Ledoit–Wolf shrinkage (Ledoit and Wolf, 2004) was proposed to mitigate exactly this, and is already the default here.

Remark 13.2 (What the negative result changes, and what it does not). *It does not change the theory. With* Σ known, *the GLS weights remain minimum-variance among linear unbiased estimators (proposition 9.1); that derivation depends on no simulation. It changes the practical recommendation: at the design sizes this paper recommends (21–42 pulls, a single year), no detectable gain in point accuracy should be expected from the covariance adjustment, and a simple mean is an equally defensible primary choice.*

It also does not change the diagnostic *value of the adjustment. The spectral effective pull count fell from 4.93 to 2.95, correctly telling the researcher that nine pulls are now worth about three. A simple mean cannot supply that judgment, nor the near-duplicate dependence structure, the per-day conditional standard errors, or measurement uncertainty in a form the downstream EIV correction can consume. RMSE cannot answer whether a batch is usable at all, and that is what this frame-*

work is mainly for.

Remark 13.3 (Reading the bias column). *Mean bias for the single pull and for the simple mean sits at the* 10^{-15} *level. That is an identity, not an estimator property: every complete pull is normalized to baseline mean 100 and the truth is defined on the same baseline, so their average cannot be biased.* RACER*-GT now lands at machine precision as well (*$+3.2\times10^{-17}$*). The cross-pull median's* -0.1464 *is real: a daily median sits systematically below the mean under the right-skewed event spikes in this DGP.*

14 Scope of the statistical guarantees

We state precisely what is and is not established.

1. Under assumptions 3.1 and 3.2, the stratified design-cell average is design-unbiased for the prespecified retrieval-mixture expectation (proposition 8.1). This holds *without* any independence assumption across pulls.
2. Under assumptions 3.3 to 3.5, the relative log-scale parameters are identified and the graph WLS estimator is conditionally unbiased and efficient among linear estimators built on the edge estimates (propositions 5.1 and 5.2).
3. Under assumption 3.6 with $\mathbf{\Sigma}$ known or consistently estimated, the unconstrained GLS weights are minimum-variance among linear unbiased estimators (proposition 9.1). The constrained default is a stabilized convex combination and does not inherit this optimality.
4. Given a correct lower-frequency benchmark and an identified constraint set, the benchmark estimator is the unique minimizer of an explicit quadratic objective (proposition 10.1).
5. Under classical additive measurement error with a consistent estimate of Ω_u, (14) corrects attenuation bias.

None of these results establishes unbiasedness with respect to Google's absolute search counts, and none is claimed to hold unconditionally.

14.1 Known limitations

- The variance components use balanced method-of-moments estimation. REML for unbalanced designs is a planned extension.

- Consensus confidence intervals are conditional approximations. A full-pipeline moving-block bootstrap—resampling historical blocks and re-estimating edges, scales, duplicates, covariance, weights, and the benchmark before re-estimating the financial model—is the appropriate procedure for confirmatory inference.
- Benchmark noise, scale-calibration uncertainty, and tuning-parameter uncertainty are not yet fully propagated into the reported intervals.
- Zero values may mix genuine zero search, privacy thresholds, sampling, and rounding. Version 1.4.0 retains zeros and isolates detection reliability, but does not implement an interval-censored latent-count likelihood. For high zero-share terms, moving to weekly frequency or to a broader Topic is usually more defensible than manufacturing daily values.
- Anchor-bank calibration across different queries is documented as an extension but is not implemented in version 1.4.0.
- Real-data validation covers the calibration stage only. Section 5.5 uses a single pull, so every mechanism from Stage 2 onward — consensus weighting, duplicate diagnosis, variance components, reliability coefficients, benchmarking — is *entirely untested* on real Google Trends data and rests on simulation alone. Table 5 has already shown the simulator overstating chunk noise by about a factor of 4.5 in the one stage where a comparison was possible, which makes it an open and consequential question whether its other parameters are distorted the same way: if cross-pull retrieval noise, day effects, or stream effects are also overstated, the relative performance figures in section 13 need re-examination too. We do not claim that simulation evidence extrapolates to cross-pull behaviour on real data. Closing the gap requires repeated collection across collection days and stream environments; under the recommended protocol, three streams by seven collection days is 21 pulls and 168 downloads, which is a month of calendar time.
- The package is deliberately ingestion-first and embeds no unofficial scraper, because endpoints, rate limits, authentication, and terms of service change. Collectors need only emit the documented long-format schema.
- Retrieval rate limiting is a binding constraint, and it undermines the design rather than merely slowing it. The confirmatory protocol recommended above — from 2010, 180-day windows, 15–30 day steps, 21–42 pulls — implies a request budget that was never stated. At a 15-day step a single pull spans

393 chunks, and each chunk costs two requests (explore for a token, then widgetdata), so one pull is 786 requests, 21 pulls is $16{,}506$, and 42 pulls is $33{,}012$. At a conservative 20-second spacing those are roughly 92 and 183 hours of continuous retrieval; even one pull takes about 4.4 hours.

Google Trends rate limits this aggressively. In one test the explore endpoint returned 200 while the *second* widgetdata request returned HTTP 429, and a retry 30 seconds later returned 429 again. A single observation cannot quantify Google's policy, which may vary with address, time of day, account state, and request pattern, but it is enough to establish that acquisition feasibility is not a detail that can be left to the collector.

The limit is not confined to programmatic access. Loading the Trends web page in an ordinary signed-in browser during the same period, by hand, produced a 429 on the front end's *own* `widgetdata` request while the chart rendered normally from cache. That has an immediate operational consequence: the Trends front end masks failed requests with cached content, so a user cannot tell from the screen whether the series shown was actually retrieved in this session. Any protocol that relies on manual export must record the response status and `raw_response_hash` independently rather than treating a rendered chart as evidence of successful retrieval. This sharpens the reason the adapter contract insists on preserving the raw response.

The damage is not the waiting. Section 4 defines `collection_day` as the ordinal level of one complete historical reconstruction, and both the D facet of the G-study and σ^2_{TD} rest on that level. If one complete reconstruction takes 4.4 hours and is repeatedly interrupted by throttling, which collection day a pull belongs to stops being well defined; a pull that crosses midnight additionally straddles the cache reset described by Djorno et al. (2026) (remark 2.2), so its chunks come from two different sampling batches. That is not measurement error, it is the failure of a facet definition.

The remedy is to size the window against the limit rather than the other way round. `examples/collect_google_trends.py --preview` prints the request budget before any data are collected, and a single year with a 180-day window and a 30-day step needs 8 chunks and 16 requests per pull, completing in minutes — short enough to keep the collection day well defined. A 16.6-year window at a 15-day step is not currently feasible, and earlier versions of this paper did not say so.

15 Conclusion

Google Trends is a measurement instrument with a documented tendency to return different numbers for the same question. Treating its output as an error-free observation is not a conservative simplification; it silently imports scale error, dependence, and attenuation into whatever model consumes it.

RACER-GT takes the opposite stance. It fixes the retrieval design before any data are seen, calibrates all chunks simultaneously on the overlap graph, measures rather than assumes the independence of repeated retrievals, decomposes the remaining variation across the facets of the design, and carries the resulting uncertainty forward into downstream estimation. What it delivers is not certainty about Google's internal counts, which is unavailable, but a reproducible, auditable, and explicitly conditional estimate of a well-defined target, together with a locked rule for deciding whether a given batch of data is fit to use at all.

The controlled evidence is deliberately mixed. Under this data-generating process, cross-pull aggregation and temporal benchmarking both deliver large and robust reductions in reconstruction error, while covariance-adjusted weighting does not beat a simple mean. We report the comparison that does not favour us alongside the ones that do, because that is the same discipline the framework asks of its users: a measurement procedure that can only produce flattering numbers cannot tell a researcher when to stop trusting it.

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